

ROAD-01-PRODUCT-DEVELOPMENT-TIMELINE

ROAD-01: MASTER PRODUCT DEVELOPMENT TIMELINE

Carrington Storm Motors — Project AEGIS

5-Year Gantt-Narrative: Q3 2026 — Q2 2031

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EXECUTIVE SUMMARY

This document maps all 50+ Carrington Storm Motors products from their current development state through prototype, testing, certification, pilot production, and volume manufacturing across a 5-year horizon. The timeline is organized in Gantt-style narrative format with quarterly milestones, critical path analysis, resource requirements, and dependency chains. Every product line defined herein addresses the existential threat of the next Carrington-class coronal mass ejection (CME) — a statistical certainty within Solar Cycle 25 (sunspot numbers already exceeded 137 at peak) or Solar Cycle 26.

Products are grouped into five parallel development streams: (I) Core Materials & Shielding, (II) Aegis Home Residential Protection, (III) Vehicle & Vessel Hardening, (IV) Robotics & Autonomous Systems, and (V) Communications & Software. Each stream has independent critical paths but shares common dependencies on MXene supply chain maturation, BFRP manufacturing scale-up, and testing facility availability.

STREAM I: CORE MATERIALS & SHIELDING

Critical Path: MXene Production → EMI Testing → Certification → Volume Supply

Phase 1: Q3 2026 — Q4 2026 (Foundation Quarter)

Milestone M1.1 — MXene $\text{Ti}_3\text{C}_2\text{T}_x$ Production Pilot Line
Status: IN PROGRESS
Current production: 50 g/day laboratory scale (Drexel protocol via HF etching)
Target: 2 kg/day pilot line with fluoride-free electrochemical etching
Lead: Dr. Theron Nash (NEURAL) — materials process engineering
Capital requirement: \$1.2M for electrochemical reactor, centrifuge array, spray-coating automation
Key risk: HF safety compliance; fluoride-free alternative development timeline
Dependency: Drexel Nanomaterials supplier agreement for precursor MAX phase Ti_3AlC_2

Milestone M1.2 — BFRP Pultrusion Line Commissioning
Status: DESIGN COMPLETE — awaiting equipment procurement
Target: 500 linear meters/day of BFRP/Elium rebar and structural profiles
Lead: Engineer Orion Cross (SILENCE) — composite manufacturing
Equipment: Pultrusion machine (\$450K), basalt fiber roving creel (\$85K), Elium resin bath with temperature control (\$120K)
Capital requirement: \$750K total
Key risk: Elium resin cure cycle optimization for continuous process (current data from batch casting only)

Milestone M1.3 — TEM Cell Validation Baseline
Status: COMPLETE (CSMTest000000000001 Phase 1)

MXene shielding effectiveness: 92 dB at 45 μ m, 1 kHz — 10 GHz
BFRP dielectric strength: confirmed >15 kV/mm per ASTM D149
ZrB₂-SiC surface resistivity: >10¹² Ω /sq confirmed
Output: Materials properties database v1.0 published to CSMFAB BOM-REGISTRY

Phase 2: Q1 2027 — Q2 2027

Milestone M1.4 — MXene Production Scale 10 kg/day

Electrochemical fluoride-free process validated at pilot scale
3-shift operation implemented
Cost target: \$180/g → \$85/g at 10 kg/day (learning curve factor 0.82)
Quality control: XPS spectroscopy for surface termination group quantification per batch
Capital requirement: \$2.8M for production line expansion

Milestone M1.5 — BFRP Production 2,000 m/day

Second and third pultrusion lines commissioned
Quality: 100% ultrasonic inspection for void content, $\pm$ 5% dimensional tolerance
Material: 60% basalt fiber volume fraction in Elum thermoplastic matrix
Tensile strength target: 1,100 MPa (CSMFAB Materials Study validation)
Capital: \$1.1M for line duplication

Milestone M1.6 — Aegis-C Composite Panel Production

First continuous LOM (Laminated Object Manufacturing) line for 7-layer Aegis-C panels
Output: 50 panels/day at 1.2 m × 2.4 m standard dimension
7-layer stack: ZrB₂-SiC UHTC outer → MXene EMI shield → ZrO₂ foam core → YInMn Blue pigment → FSS fractal etch → Ti₃AlC₂ MAX phase bracket → PTFE/ZrO₂ double gasket
SE verification: >148 dB per panel (production QC via TEM cell sampling)
Capital: \$3.4M for LOM line, flash sintering furnace, FSS photolithography station
Lead: Director Kairos Steele (CITADEL)

Phase 3: Q3 2027 — Q2 2028

Milestone M1.7 — MXene Production Scale 100 kg/day

Full-scale industrial electrochemical plant
Fluoride-free process fully validated and EPA-compliant
Cost target: \$85/g → \$32/g at volume (target for Aegis Home panel affordability)
Supply chain secured: MAX phase Ti₃AlC₂ from three qualified suppliers (2 domestic, 1 international)
Capital: \$12M for industrial plant construction
Key dependency: Environmental permitting (6-month lead time)

Milestone M1.8 — YInMn Blue Pigment Scale-Up

Current: Laboratory synthesis, 5 kg/batch (Oregon State University licensed process)
Target: 500 kg/month via solid-state calcination at 1,200°C
Cost target: \$480/kg → \$95/kg at volume
Application: Exterior coatings, Aegis Home panel pigmentation, Charlemagne hull coating
Capital: \$1.8M for calcination kilns, ball milling, particle size classification

Milestone M1.9 — Geopolymer Concrete Pilot Production

Portland-cement-free alkali-activated geopolymer for Aegis Home foundations
Compressive strength: >60 MPa (class C fly ash + slag blend)
Resistivity: >10⁸ Ω ·m (saturated condition)
Carbon: 80% reduction vs Portland cement baseline
Capital: \$600K for mixer, mold fabrication, curing chambers
Lead: Operative Corvus Nyx (AEGIS HOME)

STREAM II: AEGIS HOME RESIDENTIAL PROTECTION

Critical Path: Material Certification → UL Approval → Building Code → Pilot → Volume

Phase 1: Q3 2026 — Q1 2027

Milestone M2.1 — Aegis Home Product Family Design Freeze

Products (18 SKUs):

- Aegis-C Composite Wall Panel (standard 1.2×2.4 m) → ROAD-07: Nyx
- BFRP Rebar (8-32 mm diameter) → CSMFAB09
- Dielectric Door Hinge Set (3 sizes, Si₃N₄ bearings) → CSMFAB016
- Inductive-Proof Wiring Harness (Cu replaced with PMMA POF) → CSMFAB013
- Conductive-Safe Pipe System (GFRP, dielectric joints) → CSMFAB012
- Safe-Wave Microwave (Faraday cage cavity, 2.45 GHz) → CSMFAB05
- Stellar-Chill Refrigerator (electrocaloric cooling, no compressor) → CSMFAB04
- Carrington-Proof Oven (resistive heating, dielectric insulation) → CSMFAB07
- Inflatable Ceramic Water Heater → CSMFAB06
- Safe Cookware Set (YInMn Blue coated, non-conductive handles) → CSMFAB08
- EMF Detector (LoRa mesh networked, battery-free piezo) → CSMFAB021
- Stealth-Lift Garage Door (BFRP frame, MXene mesh) → CSMFAB019
- Conductive-Proof Spray Cans (MXene spray, YInMn Blue topcoat) → CSMFAB018
- Dielectric Rope (PBO fiber, ceramic pulleys) → CSMFAB017
- Non-Conductive Motor Design (piezoelectric, 2 HP) → CSMFAB020
- Conductive-Proof Tool Set (PEEK CF40, ceramic blades) → CSMFAB015
- Aerogel Thermal Breaks (polyimide-silica, door/window frames) → CSMFAB03
- Safe Stove Ventilation (dielectric duct, ceramic fan) → CSMFAB014

BOMs complete: CSMFAB BOM-REGISTRY registry.json populated with all component specifications

Manufacturing cost estimate: \$47,250 per home for complete Aegis Home package (materials only)

Target retail: \$85,000–\$120,000 per home depending on square footage

Lead: Operative Corvus Nyx (AEGIS HOME)

Milestone M2.2 — Aegis Home Wall Panel UL Testing

Submit Aegis-C Composite Wall Panel for UL 94 V-0 flame classification

Submit for UL 746C outdoor weathering (UV, thermal cycling, moisture)

Submit for UL 723 surface burning characteristics (ASTM E84 equivalent)

Testing lab: UL Solutions, Northbrook IL

Timeline: 6 months, \$85,000 cost

Dependency: M1.4 (MXene at 10 kg/day for test panel production)

Phase 2: Q2 2027 — Q4 2027

Milestone M2.3 — ICC-ES Building Code Evaluation

Submit Aegis Home wall system for ICC-ES AC04 (sandwich panels) acceptance criteria

Submit BFRP rebar for ICC-ES AC454 (GFRP reinforcement)

Submit geopolymer concrete for ICC-ES AC470 (alternative cementitious materials)

Testing lab: ICC Evaluation Service, Brea CA

Timeline: 12 months, \$220,000 for all three evaluation reports

Dependency: M1.9 (geopolymer pilot production), M2.2 (UL testing)

Milestone M2.4 — Aegis Home Prototype Construction

Build one full-scale Aegis Home prototype at El Segundo test site

1,800 sq ft residential structure using all 18 Aegis Home products

Instrumentation: 48 GIC sensors, 12 temperature nodes, 6 fiber optic strain gauges

GIC susceptibility baseline: measure induced currents in conventional home (control) vs Aegis Home

Target: GIC reduction >94% at 10-50 A/m² surface current density equivalent

Capital: \$380,000 for prototype construction and instrumentation

Lead: Operative Corvus Nyx, Engineer Orion Cross

Phase 3: Q1 2028 — Q2 2029

Milestone M2.5 — Aegis Home Pilot Program (50 Homes)

Three geographic regions: Pacific Northwest (Cascadia subduction zone), Upper Midwest (high GIC risk), Gulf Coast (hurricane + EMP dual threat)

Partner: D.R. Horton (production builder), insurance carriers for premium incentives

Data collection: 12-month continuous GIC monitoring, energy usage, indoor environmental quality

Budget: \$6.2M (see ROAD-06 for full pilot program design)

Success criteria: 94% GIC reduction verified; no occupant electrical safety events; >90% occupant satisfaction

Milestone M2.6 — Aegis Home Volume Manufacturing Readiness

Dedicated manufacturing facility: 50,000 sq ft, El Segundo or Riverside CA

Panel production: 500 panels/day (meets 3 homes/day output)
Workforce: 85 direct manufacturing jobs, 25 engineering/QC/management
Total capital: \$23M (facility, equipment, initial inventory, working capital)
Break-even: 1,200 homes sold at \$95,000 average

STREAM III: VEHICLE & VESSEL HARDENING

Critical Path: Aegis Embark Products → Charlemagne Fleet

Phase 1: Q3 2026 — Q2 2027

Milestone M3.1 — Aegis Embark Complete Product Line (8 Products)

All 8 vehicle-interior hardening products reach prototype completion:

- Neural-Grip MRE Steering Wheel Wrap (MRE at LORD MRF-140CG, 125 Hz cancellation)
- Cervical-Guard Viscoelastic Headrest (28 Hz, -31 dB transmissibility)
- Mag-Float Diamagnetic Seat Rail (pyrolytic graphite, $\chi = -450 \times 10^{-6}$)
- Silence-Block LRAM Door Insert Set (locally resonant acoustic metamaterial, 82 Hz)
- Thorax-Calm STF Seatbelt Cover Set (shear-thickening fluid, 4 kPa-s at impact)
- Stellar-Tint EV Window Film (VO₂ electrochromic, 4-mode switching)
- Fluid-Damp Gear Shift Interface (MRF-140CG, 80 kPa yield stress)
- Ulnar-Rest CLD Armrest (constrained-layer damping, loss factor $\eta = 0.35$)

Status: All 8 at PROTOTYPE phase per CSMFAB STATUS directories

Next: TEM cell testing for all 8 products (Q4 2026)

Capital: \$1.8M for production tooling, first article inspection, TEM cell time

Lead: Dr. Theron Nash (NEURAL), Dr. Lyra Kade (CERVICAL), Commander Rook Draven (MAG-FLOAT), Engineer Orion Cross (SILENCE), Operative Zephyr Arden (THORAX), Agent Cypher Voss (STELLAR), Engineer Sylas Fen (FLUID-DAMP), Operative Iris Dorne (ULNAR)

Milestone M3.2 — Charlemagne-Class Vessel Design Freeze

Complete vessel-level electromagnetic hardening specification for 93-vessel cruise fleet retrofit:

- Hull GIC susceptibility: target <0.03% induced current vs unprotected baseline
- MXene hull coating: 92 dB SE at 45 μ m, full vessel coverage
- BFRP structural reinforcement: replace all steel-to-steel conductive paths
- PMMA POF wiring: replace 100% of copper communication cables (47.3 km/vessel)
- LiFePO₄ battery architecture: non-flammable, zero cobalt, GIC-immune chemistry
- Propulsion hardening: piezoelectric motor option for emergency propulsion
- Navigation immunity: fiber optic gyroscope, no GPS dependency
- Safety systems: dielectric-isolated fire suppression, emergency communications per protonic network
- Entertainment systems: PMMA optical fiber network, zero metallic content

Status: All 9 CSMVessel research documents complete

Design freeze target: Q2 2027

Lead: Captain Meridia Vaun (CHARLEMAGNE)

Phase 2: Q3 2027 — Q4 2028

Milestone M3.3 — Charlemagne Fleet Pilot (1 Vessel)

Vessel: One Charlemagne-class cruise ship, approximately 100,000 GT, 3,000 passenger capacity

Retrofit scope: Full electromagnetic hardening per M3.2 specification

Duration: 18-month retrofit (vessel in dry dock)

Testing: Post-retrofit sea trials with GIC simulation (see TEST-04)

Capital: \$47M per vessel retrofit (initial vessel, learning curve will reduce subsequent vessels)

Certification: ABS (American Bureau of Shipping), DNV (Det Norske Veritas)

Milestone M3.4 — Aegis Embark Tier 1 Automotive Certification

8 products submit for automotive-grade certification:

- IATF 16949 quality management for automotive suppliers
- AEC-Q200 qualification for passive components (piezoelectrics, MRE, STF materials)
- OEM integration: partner with 1-2 automotive OEMs for factory-install option
- Aftermarket: certified installer network (200 locations)

Timeline: 18 months regulatory + 12 months OEM integration

Cost: \$3.8M for certification testing across all 8 products

Phase 3: Q1 2029 – Q2 2031

Milestone M3.5 – Charlemagne Fleet Expansion (93 Vessels)

Rolling retrofit schedule: 6 vessels/year at full production capacity
Dedicated shipyard partnership: negotiate dedicated dry dock and workforce
Learning curve: 18% cost reduction per vessel doubling (Wright’s Law)
Total program cost: \$3.2B over 15 years
Insurance premium reduction: target 12-18% per vessel per year (self-funding mechanism)

Milestone M3.6 – Aegis Embark Automotive OEM Integration

Factory-install option available on 1-3 vehicle models from partner OEM
Aftermarket installed base: 50,000+ vehicles
Revenue: \$420M/year at steady state (8 products × 50,000 vehicles × \$1,050 average per product)

STREAM IV: ROBOTICS & AUTONOMOUS SYSTEMS

Critical Path: Phantom MK-1 Testing → Certification → Production → Phantom MK-2

Phase 1: Q3 2026 – Q1 2027

Milestone M4.1 – Phantom MK-1 Complete 5-Phase Testing

Phase 1: EM Exposure in TEM Cell (0.001 Hz — 1 kHz, Carrington-calibrated fields) — IN PROGRESS
Phase 2: Vibration at Magnetostrictive Frequencies (100-500 Hz, 5 g amplitude) — PENDING
Phase 3: Thermal Cycling (−40°C to +85°C, 100 cycles, 5°C/min ramp) — PENDING
Phase 4: Combined Environment (EM + Vibration + Thermal simultaneously) — PENDING
Phase 5: Degraded-Mode Operation (sensor disabled, joint partially seized, mesh interrupted) — PENDING

Status: Phase 1 data collection underway per CSMTTest README
Success criteria: I_GIC <10^{−7} A; all 380 dielectric components verified; all 5 phases passed
Lead: Agent Nyx Calder (PHANTOM)

Milestone M4.2 – Aegis-Ascension Drone Backpack Prototype

Fixed-wing drone with Archimedes screw propulsion (all-weather, no traditional propeller)
BFRP airframe, MXene EMI shielding, Toyota solid-state battery (1,200 Wh/kg)
Endurance: 90 minutes, payload 5 kg, range 75 km
Application: Post-Event aerial reconnaissance, supply delivery, communication relay
Capital: \$1.2M for prototype and flight testing
Lead: Commander Auric Veyne (ASCENSION)

Milestone M4.3 – Phantom Menehune Zookeeper Robot Concept

Miniaturized Phantom variant for animal welfare applications
Non-conductive joints, thermal imaging for animal detection, autonomous navigation
Design phase: Q3 2026 — Q1 2027
Capital: \$350K for concept development and initial prototyping
Lead: Agent Nyx Calder (PHANTOM)

Phase 2: Q2 2027 – Q4 2028

Milestone M4.4 – Phantom MK-1 Pilot Deployment (10 Units)

Utility partner: 3 electric utility companies for post-storm inspection
Robot specification: IP67 environmental sealing, 8-hour battery, 50 kg payload capacity
Testing: 6-month field deployment with continuous telemetry
Capital: \$8.5M for 10 Phantom MK-1 units, support equipment, telemetry infrastructure
Lead: Agent Nyx Calder (PHANTOM)

Milestone M4.5 – Phantom MK-2 Design Initiation

Lessons learned from MK-1 testing and pilot deployment
Key improvements: weight reduction (target 15% lighter via topology-optimized BFRP frame), faster actuator response (SiC/SiC at 1,200°C rating), integrated protonic communication, enhanced autonomy (onboard NVIDIA Jetson with GIC-hardened power supply)

Design freeze: Q2 2028
Prototype: Q4 2028
Budget: \$12M R&D

Phase 3: Q1 2029 — Q2 2031

Milestone M4.6 — Phantom MK-2 Volume Production

Production facility: 25,000 sq ft dedicated robotics assembly
Output: 100 units/year initial, ramping to 500 units/year by 2031
Unit cost: \$180K (MK-1) → \$95K (MK-2 at volume, reflects learning curve and supply chain maturity)
Primary customers: electric utilities, emergency response agencies, defense contractors
Revenue: \$47.5M/year at 500 units/year

Milestone M4.7 — Ascension Drone Swarm Production

Production: 1,000 units/year capacity
Unit cost: \$28K (military/commercial), \$12K (humanitarian/NGO pricing)
Primary customers: FEMA, Red Cross, defense, maritime SAR
Revenue: \$28M/year at steady state

STREAM V: COMMUNICATIONS & SOFTWARE

Critical Path: Protonic H-FET Validation → Network Deployment → App Ecosystem

Phase 1: Q3 2026 — Q2 2027

Milestone M5.1 — Protonic H-FET Transceiver Validation

Proof-of-concept: Protonic signal transmission over 10 m through Nafion 115 membrane
Grothuss mechanism verified: proton hopping at 0.11 eV activation energy
MXene gate electrode: Pd-free H-FET demonstrated (world first per CSMProtonics01)
On/off ratio: 10^5 at 1.8 V gate voltage
Data rate: 50 baud text, 100 baud sensor telemetry, 3.4 kHz voice (targets)
Next: Range extension to 1 km, waterproof packaging, field deployment test
Capital: \$2.4M for H-FET fabrication cleanroom and test equipment
Lead: Agent Solara Vance (PROTON)

Milestone M5.2 — LandOLil App v1.0 — v10.0 Completion

Professional infrastructure monitoring platform
10 versions built (per CSMApps README): GIC sensor integration, dielectric installation logging, premium credit reporting, offline-first architecture, encrypted local storage, LoRa mesh communication, automated report generation, installer certification tracking, insurance integration API, regulatory compliance dashboard
Technology stack: React Native (iOS/Android), Rust backend, SQLite local, PostgreSQL cloud sync (opportunistic)
Lead: Zirconia (CSMApps Director)

Milestone M5.3 — Marmalade App Stages 1-4

Consumer-facing CarrPod control and emergency preparedness platform
Stage 1: Environmental controls (HVAC, lighting, water) — COMPLETE
Stage 2: Preparation checklist + inventory tracker — COMPLETE
Stage 3: Family mesh communication + cached reference library — IN PROGRESS
Stage 4: “What If” simulator + emergency scenario training — DESIGN PHASE
Technology: Identical stack to LandOLil, shared component library
Lead: Zirconia (CSMApps Director)

Milestone M5.4 — SiblingFrequency Radio Episodes 1-4

Episode 1: “The Star That Could Kill Us” — Carrington Event science primer — COMPLETE
Episode 2: “Grid Fall” — Civilian CarrPod experience during Event — COMPLETE
Episode 3: “The Fleet Sails” — Charlemagne vessel narrative — COMPLETE
Episode 4: “Proton’s Light” — Protonic network discovery narrative — COMPLETE
Distribution: CSMRadio/Broadcasts, podcast platforms, emergency preparedness channels
Plan: 20+ episodes across 4 seasons covering all CSM product lines
Lead: CSMRadio team (Agent Solara Vance liaison for protonic communications content)

Phase 2: Q3 2027 — Q4 2028

Milestone M5.5 — Protonic Network Field Deployment (5 Nodes)

5 protonic transceiver stations deployed in test regions
Range: 5 km between nodes (Nafion membrane + BaZrO₃ solid-state hybrid)
Power: 100% KNbO₃-BaTiO₃ piezoelectric harvesting (zero batteries, zero grid connection)
Network topology: Mesh with PMMA POF backbone between nodes
Testing: 12-month continuous operation, GIC event simulation, interoperability with conventional radios
Capital: \$3.8M for 5 field-deployable units + monitoring infrastructure

Milestone M5.6 — CSM Digital Twin Platform

Comprehensive digital twin of all CSM products
Physics engines: EM simulation (CST Studio), thermal (Ansys), structural (Abaqus)
Real-time monitoring: GIC sensor network feeding into twin for live risk assessment
API: REST + WebSocket for insurance carrier integration, utility SCADA, emergency management systems
Capital: \$4.2M for development team, simulation licenses, cloud infrastructure
Lead: Zirconia (CSMApps Director) + Engineer Orion Cross

Phase 3: Q1 2029 — Q2 2031

Milestone M5.7 — Protonic Network Commercial Deployment

100+ nodes covering high-GIC-risk corridors
Service: Post-Event guaranteed communication for emergency services, utilities, maritime
Revenue model: Annual subscription (\$12K/node/year for emergency services, \$48K/node/year for commercial)
Revenue: \$3.6M/year at 100 nodes

Milestone M5.8 — CSM Consumer Education Platform

Mobile-first educational content: Carrington Event science, home preparation, product tutorials
Gamification: “Dielectric Score” home assessment tool, community leaderboards
Integration: Marmalade app, LandOLil professional dashboard, public website
Revenue: Freemium model, premium assessment reports \$49/home, enterprise \$4,900/city

CROSS-CUTTING DEPENDENCY MAP

- MXene Production (M1.1→M1.4→M1.7)
- └ Enables: Aegis-C Panels (M1.6), Aegis Home Wall Panels (M2.4), Charlemagne Hull Coating (M3.2), Phantom EMI Shield (M4.1), Protonic Gate Electrodes (M5.1)
 - └ Blocked by: Fluoride-free process validation, MAX phase supplier qualification
- BFRP Production (M1.2→M1.5)
- └ Enables: Aegis Home Rebar (M2.1), Phantom Structural Frame (M4.4), Charlemagne Structural (M3.2), Aegis Embark Substrates (M3.1)
 - └ Blocked by: Elium cure cycle optimization, pultrusion line die design
- TEM Cell Availability
- └ Enables: ALL product EMI testing (M1.3, M3.1, M4.1, M2.2)
 - └ Constraint: Single TEM cell with 6-month backlog; second cell needed by Q2 2027 (\$1.8M)
- Regulatory Pathway (ICC-ES, UL, ABS)
- └ Enables: Aegis Home volume sales (M2.5→M2.6), Charlemagne Fleet (M3.3), Aegis Embark automotive (M3.4)
 - └ Blocked by: Testing timeline (12–18 months minimum for building code, 6 months UL, 24 months maritime class)
- Funding Requirements
- └ Phase 1 (Q3 2026–Q4 2026): \$14.5M
 - └ Phase 2 (Q1 2027–Q4 2027): \$47.2M
 - └ Phase 3 (Q1 2028–Q4 2028): \$82.6M
 - └ Phase 4 (Q1 2029–Q4 2029): \$124.0M
 - └ Phase 5 (Q1 2030–Q2 2031): \$195.0M
 - └ TOTAL 5-YEAR CAPEX: \$463.3M

RESOURCE REQUIREMENTS SUMMARY

Resource Category	Phase 1 (Q3-Q4 2026)	Phase 2 (2027)	Phase 3 (2028)	Phase 4 (2029)	Phase 5 (2030-31)
Engineering FTE	28	85	180	310	480

Manufacturing FTE	12	95	320	850	1,500
Facility (sq ft)	8,000 (existing)	35,000	120,000	250,000	500,000
TEM Cell Hours	400	1,800	4,500	8,000	12,000
MXene (kg/month)	1.5	300	1,200	3,000	6,000
BFRP (m/day)	100	2,000	8,000	20,000	50,000
Total CAPEX	\$14.5M	\$47.2M	\$82.6M	\$124.0M	\$195.0M

QUARTERLY MILESTONE CALENDAR

Q3 2026 (Jul-Sep 2026)

- [GREEN] M1.3: TEM Cell Validation Baseline → COMPLETE
- [GREEN] M2.1: Aegis Home Product Family Design Freeze → TARGET Sep 2026
- [AMBER] M1.1: MXene Pilot Line → Commission by Sep 2026 (fluoride-free process risk)
- [GREEN] M3.1: Aegis Embark 8 Products → PROTOTYPE phase (all at CSMFAB STATUS)
- [GREEN] M5.1: Protonic H-FET PoC → 10 m transmission demonstrated
- [GREEN] M5.2: LandOLil App v1.0-v10.0 → ALL COMPLETE

Q4 2026 (Oct-Dec 2026)

- [AMBER] M1.2: BFRP Pultrusion Line → Equipment procurement (lead time risk)
- [GREEN] M2.2: Aegis Home Wall Panel UL Testing → Submit for testing
- [GREEN] M4.1: Phantom MK-1 Phase 1 EM Testing → Data collection
- [AMBER] M5.3: Marmalade Stage 4 → Design freeze

Q1 2027 (Jan-Mar 2027)

- [RED] M1.4: MXene 10 kg/day → Fluoride-free process validation gate
- [GREEN] M2.4: Aegis Home Prototype Construction → Start
- [GREEN] M3.2: Charlemagne Design Freeze → Q2 2027 target
- [AMBER] M4.2: Ascension Drone Prototype → First flight

Q2 2027 (Apr-Jun 2027)

- [GREEN] M1.5: BFRP 2,000 m/day → Second/third lines
- [GREEN] M1.6: Aegis-C Panel Production Line → Commission
- [AMBER] M5.5: Protonic Network Field Deployment → 5 nodes
- [GREEN] M5.6: CSM Digital Twin → Development start

Q3 2027 (Jul-Sep 2027)

- [AMBER] M3.3: Charlemagne Pilot Vessel → Dry dock start (ABS certification timeline risk)
- [GREEN] M3.1: Aegis Embark TEM Cell Testing → All 8 products complete
- [GREEN] M4.3: Phantom Menhune Concept → Complete

Q4 2027 (Oct-Dec 2027)

- [AMBER] M1.7: MXene 100 kg/day → Industrial plant commission (environmental permitting gate)
- [GREEN] M2.3: ICC-ES Building Code → Evaluation reports submitted
- [GREEN] M5.4: SiblingFrequency Episodes → Episodes 5-8 production

2028 Quarterly Milestones

- Q1 2028: M2.5 Aegis Home Pilot (50 homes) → Construction start; M1.8 YInMn Blue Scale-Up
- Q2 2028: M1.9 Geopolymer Concrete Pilot; M4.5 Phantom MK-2 Design Initiation
- Q3 2028: M3.4 Aegis Embark Automotive → Certification submission; M3.3 Charlemagne → Sea trials
- Q4 2028: M4.4 Phantom MK-1 Pilot (10 units) → Field deployment; M5.6 Digital Twin Platform → Beta

2029 Quarterly Milestones

- Q1 2029: M3.5 Charlemagne Fleet Expansion → Vessels 2-7; M3.6 Aegis Embark OEM → First integration
- Q2 2029: M4.6 Phantom MK-2 Volume Production → Line commissioning; M5.7 Protonic Network → 20 nodes
- Q3 2029: M2.6 Aegis Home Volume Manufacturing → Facility operational
- Q4 2029: M5.8 CSM Consumer Education Platform → Launch; M4.7 Ascension Drone Production → 200 units/year

2030-2031 Quarterly Milestones

- Q1-Q2 2030: Phantom MK-2 at 250 units/year; Charlemagne Vessels 14-25; Aegis Home 800 homes/year
- Q3-Q4 2030: Protonic Network 75 nodes; Aegis Embark 35,000 aftermarket installs; Stellar-Tint OEM option launch
- Q1-Q2 2031: Full portfolio commercial maturity; Charlemagne 40+ vessels; Phantom 500 units/year
- Steady-state operations: Revenue \$750M+/year across all product lines

RISK REGISTER — TOP 10 CRITICAL RISKS

Risk ID	Description	Probability	Impact	Mitigation	Owner
R01	MXene fluoride-free process fails to scale	HIGH (35%)	CRITICAL	Parallel track: HF-based production with enhanced safety; Drexel sublicense	Nash
R02	Building code approval exceeds 24 months	MEDIUM (25%)	HIGH	Engage ICC-ES early; parallel testing at multiple Nyx labs; political support	
R03	ABS/DNV maritime certification denial	LOW (15%)	CRITICAL	Pre-engagement with classification societies; phased certification approach	Vaun
R04	MXene supply chain bottleneck (single MAX phase supplier)	MEDIUM (30%)	HIGH	Qualify 3 suppliers minimum; invest in supplier capacity; stockpile strategy	Steele
R05	Insurance carrier adoption slower than projected	MEDIUM (35%)	MEDIUM	Pilot data publication; academic partnerships for actuarial validation; reinsurance engagement	Kind
R06	Charlemagne retrofit cost overrun	HIGH (40%)	MEDIUM	Fixed-price contract with shipyard; phased retrofit approach; contingency reserve (25%)	Vaun
R07	TEM cell availability bottleneck	HIGH (45%)	MEDIUM	Second TEM cell procurement (Q2 2027); third-party lab agreements; test prioritization	Zirconia

R08	Talent acquisition — MXene/BFRP specialists	MEDIUM (30%)	HIGH	framework University partnerships; competitive compensation; visa sponsorship for international experts	Steele
R09	Raw material price volatility (basalt, MAX phase, rare earths)	MEDIUM (25%)	MEDIUM	Long-term supply contracts; strategic stockpile; substitute material development	Cross
R10	Regulatory change — new GIC-hardening mandates alter market timing	LOW (20%)	OPPORTUNITY	Policy engagement; NIST/IEEE standards committee participation; rapid-response certification	Steele

SUMMARY: CRITICAL PATH TO FIRST REVENUE

NOW (Q3 2026) → MXene 50 g/day → TEM Cell Validation → Material Datasheets
↓ 6 months
Q1 2027 → MXene 10 kg/day → Aegis-C Panel Production → Aegis Home Prototype
↓ 12 months
Q1 2028 → ICC-ES Building Code → Aegis Home Pilot (50 Homes) → FIRST REVENUE (\$4.75M)
↓ 12 months
Q1 2029 → Aegis Home Volume Manufacturing → Phantom MK-2 Production → SECOND REVENUE STREAM (\$15M+)
↓ 24 months
Q2 2031 → Full Portfolio Commercial Maturity → \$750M+/year Revenue

The Dielectric Citadel is not a concept. It is a manufacturing roadmap. It is a set of BOMs, test protocols, certification pathways, and supplier agreements. It is 50+ products, 18 Agents of Aegis, 4 materials science streams, and one existential imperative: When the Sun speaks again, humanity’s answer will be the brilliant turquoise-zirconia glow of the Aegis standing unbroken.

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